CHE 37800: Heat and Mass Transfer

Spring 2026 (4 credits)
Course Information

Davidson School of Chemical Engineering
Purdue University

Course Goals

Develop a sound fundamental understanding of heat and mass transfer through theoretical analyses and consideration of the physical phenomena. Integrate the concepts of momentum, heat and mass transfer to acquire an understanding of the interrelation of these physical phenomena. Apply the concepts to solve problems of practical importance in chemical engineering and allied fields. Continue the development of problem-solving and self-learning skills.

A majority of the concepts presented in this course form the basis for a large number of chemical engineering unit operations, some of which are developed further in, for example, CHE 435 (Chemical Engineering Laboratory). Thus, the material covered in this course is an essential component of the chemical engineering practice and is essential that you have a clear understanding of this material.

Prerequisites

The prerequisites are CHE 211 and 377. You must have passing grades in these courses to remain enrolled in CHE 378.

Textbooks

See "Textbook Info" in Brightspace Content to find books on reserve through Purdue Libraries.

Required

S. Middleman, *An Introduction to Mass and Heat Transfer*, John Wiley & Sons, Inc., 1998.

Supplemental Texts

Y.A. Çengel *et al.*, *Heat and Mass Transfer: Fundamentals & Applications*, 5th edition, McGraw-Hill, 2015.

Incropera *et al.*, *Fundamentals of Heat and Mass Transfer*, 6th edition, John Wiley & Sons, Inc., 2007.

Welty *et al.*, *Fundamentals of Momentum, Heat, and Mass Transfer*, 5th edition, John Wiley & Sons, Inc., 2008.

E.L. Cussler, *Diffusion: Mass Transfer in Fluid Systems*, 3rd edition, Cambridge University Press, 2009.

Website for Course Information

This course will use the Purdue Brightspace site. The website is limited to enrolled students and will have the syllabus, handouts, homework, and other important class information. To login, use your university name and password. Please check the website regularly for assignments as paper

copies will not be handed out in class. Generally, homework is due on Thursdays, and homework will be posted a week before it is due. Recitation problems will be posted prior to recitation sessions, and you should bring a copy of the problems to recitation. To get an e-mail/text notification of new announcements, follow the directions here.

E-mail

Occasionally, important class announcements will be disseminated through the class e-mail list. It is your responsibility to regularly check your e-mail every day and to read the e-mails regarding CHE 37800 to receive important class information. E-mail is the preferred mode of contact. Please put CHE 378 in your subject line. If you e-mail with questions or a request to make an appointment, please allow a minimum of 24 hours for a response during the week (or a response by Tuesday morning if the e-mail is sent on the weekend).

Homework

Homework will be submitted through Gradescope, which can be accessed through the Purdue Brightspace website. Online submissions must be done by 11:59 p.m. on the due date stated. *After this time, no late homework will be accepted*. Please include on the first page your name, the assignment number, the date, and the names of the people you worked with. You may work in groups; the sharing of ideas is an excellent way to learn. Individual solutions, however, have to be turned in by each student. Experience has shown that if you do not develop your own solution to the homework, your performance on the exams will suffer. It is possible that the whole homework assignment may be graded or that only specific problems on a homework assignment may be graded. Each homework assignment will be weighted evenly when calculating the homework portion of your grade. *The lowest homework grade will be dropped*. A student *has one week after the graded homework has been released* to submit a regrade request via Gradescope.

A major component of CHE 378 will be the use of computers to solve more difficult heat and mass transfer problems. You should have been exposed to Matlab as part of the freshman computing course, though you are free to use other software packages when numerical solution is required (such as Excel). All computer questions should be referred to the TAs.

Fundamentals Laboratory

Four experiments will be performed in the Fundamentals Laboratory (FRNY G111). They will provide hands-on experience with various concepts developed in class and will thus serve as a complement to the lecture material. Each experiment will require 1) preparation prior to the lab session, 2) work in the laboratory in a team, and 3) preparation of a brief written report. Good laboratory practices will be required in the laboratory. You will not be admitted into the laboratory without 1) safety glasses with side shields (you must provide your own), 2) long pants, and 3) closed toe shoes (i.e. no flip-flops). If you do not come properly dressed, you will NOT be admitted into the laboratory and will incur an unexcused absence for that lab project. Also, you will be permitted to bring only writing materials and a laptop into the lab (i.e. no coats, food or drink, or bookbags). A locker should be assigned to you.

Materials can be accessed at the Fundamentals Lab website. There is a quiz that needs to be completed before entering the lab.

Recitations
Recitation sections will be primarily used for group-based pre-laboratory assignments and practice problems. Graded activities will be assigned, some performed during a given weekly recitation and others due at a particular recitation session. The same team of students working together in the Fundamentals Laboratory will also be working together during the recitations. Some recitations will also be used to solve additional problems related to the course material.

Exams
The class is scheduled to have two evening exams. The final exam will be two hours and given during finals week. The date/time will be determined by the registrar during the semester.

Every exam will be comprehensive, with some possible emphasis on the more recent material. Exams will be closed-book-and-notes, unless announced otherwise. Exam dates and formats are subject to change.

Any student who cannot take an exam as scheduled (e.g., due to conflicts with another exam) *must* make special arrangements by sending an e-mail at least one month *before* the exam. Individual make-up exams *will not* be given. In cases of extenuating circumstances (e.g., illness, quarantine, bereavement) or extreme duress (e.g., hospitalization), please provide documentation, and if possible, speak to me *before* the exam takes place. Travel plans do not constitute extenuating circumstances.

A student *has one week after the graded exam has been released* to submit a regrade request via Gradescope. We will regrade the whole exam when you request a regrade; thus, your grade may go up or down.

Grades
Grades will be determined as such:

5%	Professionalism/Participation
15%	Homework
15%	Laboratory Reports, Recitation Activities
20%	Exam 1
20%	Exam 2
25%	Final Exam

For group activities (e.g., lab reports), we will use CATME software to collect peer evaluations on the degree of participation/contribution of all group members, the results of which will be used to adjust the grade you actually receive for that project.

The final grades will be assigned based primarily on the absolute performance and secondarily on the relative performance. The following grading scale is guaranteed but may be modified based on relative student performance:

A+	98%-100%	C	74-76%
A	94-97%	C-	70-73%
A-	90-93%	D+	67-69%
B+	87-89%	D	64-66%
B	84-86%	D-	60-63%
B-	80-83%	F	<60%
C+	77-79%		

Course Evaluation Policy

At the end of the semester, you will be asked to fill out an online course evaluation. It is important that you give honest and critical feedback for future course improvement. To encourage participation, extra credit will be provided to each student who submits confirmation that he/she has filled out the survey.

Professionalism

The highest standards of professionalism and ethics are expected in CHE 378. Each student is expected to come to class on time and not disrupt the class. Each student is also expected to follow Purdue's student conduct code and behave in a professional manner. The rights of students in violation of the code of conduct are outlined on Purdue's website. Each student is expected to exhibit consideration and respect towards the other students, the graders, the teaching assistants (TAs), and the professor. Each student is expected to exhibit a positive attitude. Expectations for each student include (but are not limited to):

a. Attending lecture and required recitation/laboratory sessions.
b. Coming to class, recitation, and lab on time and prepared by reading assigned material beforehand.
c. Refraining from disrupting class (e.g., silencing cell phones, refraining from cell phone or laptop use during class, and carrying on a loud conversation during class).
d. Maintaining the highest standards of academic honesty and integrity.
e. Being an active contributor to team assignments and completing your portion of the assignment in a timely manner.
f. Being knowledgeable about the policies and information described in the syllabus.

Any student who does not meet the professionalism standards will not receive full points for this portion of the grade.

Attendance Policy

This course follows the Academic Regulations: Attendance and Office of the Dean of Students: Class Absences policy posted in Brightspace under "University Policies and Statements." This policy states that students are expected to be present for every meeting of the classes in which they are enrolled. It is understood that, occasionally, you may miss lecture due to unforeseen circumstances (e.g., illness – please do not come to class if you are feeling ill), and, in these cases, you can ask any questions about missed concepts during office hours. Unless it falls under the University excused absence regulations (see below), any work due should be submitted on time. In cases falling under excused absence regulations, the student or the student's representative

should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care.

Because of assigned group projects, attendance is required in the laboratory and recitation sections. Only absences due to extreme circumstances (e.g., your own hospitalization, death in the family, or mandatory military training) will be excused, and the student should send an e-mail and provide appropriate documentation. Unexcused absences will result in a significant grade reduction for the corresponding lab. In the case of an unexcused lab absence, you may be eligible to receive up to 50% of the assigned grade for the lab report if you actively participate in the writing of the lab report.

Illness

If a student becomes sick (e.g., with flu-like symptoms), the student should seek prompt medical attention, and then not come back to class until the student has been symptom-free for more than 24 hours. That is, the student should utilize the resources at the Purdue University Student Health Center (PUSH) or another trained medical professional. If possible, the student should let me know as soon as possible such that so that we can communicate about how you can maintain your academic progress. Reasonable accommodations will be made on an individual basis to ensure that all students have an opportunity to learn. In the event of a severe outbreak of illness at Purdue that mandates that our class not meet, all attempts will be made to deliver the course online.

Academic Integrity

The highest standards of academic honesty are expected. The Purdue Honor Pledge is: "As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue." Purdue University's policy on academic dishonesty states that "the commitment of the acts of cheating, lying, stealing, and deceit in any of their diverse forms (such as the use of ghost-written papers, the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest" (University Senate Document 72-18, December 15, 1972). Plagiarism means "to use and to pass off someone else's ideas, inventions, writings, etc. as one's own" (New Webster's Dictionary and Thesaurus, 1992). You should not copy portions of the lab handout in your lab reports. Please note that for this class, you should not copy or re-use your own previously written work (e.g., do not copy a section from a lab report you wrote for another class) as it will be deemed plagiarism. You will submit your lab reports through Brightspace, and it will be checked by Turnitin for plagiarism and unauthorized use of AI. Unless specified by the instructor in writing, AI tools may only be used during the learning process to summarize readings. Other acts of dishonesty include: (i) possessing or accessing, in hardcopy or electronic form, the solutions to the course text, previous years' homework problems and exams (e.g., including obtaining answer keys from websites such as Course Hero, Chegg, etc.); (ii) claiming credit for work that is not your own original work; and (iii) enabling other students to create work that is not their original work. In this course, cheating, plagiarism, or any act of dishonesty will not be tolerated.

Any incidents of academic dishonesty will be reported to the Office of Student Rights and Responsibilities where university penalties, including removal from the university, may be considered. The first offense will result in an F on the pertinent homework assignment, recitation activity, exam, or lab report. A second offense will result in an F grade for the course.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling (765) 494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Campus Emergency

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. You are expected to check your @purdue.edu email address frequently. Here are ways to obtain information about changes in this course.
Course Brightspace Page: Spring 2026 CHE 37800-022 LEC

Emergency Procedures

In the event that the class would need to evacuate ARMS (e.g., in the event of a fire alarm), the class should proceed to exit the building and meet on the west side of ARMS (Engineering Mall area). In the case of inclement weather, we will meet in the FRNY atrium. Do not leave the area as emergency responders will need to count to ensure that all persons have made it from the facility. In the event that we are required to shelter in place, we will shelter in place in our classroom (ARMS 1010).

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let the professor know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at drc@purdue.edu or by phone: 765-494-1247.

In addition to the University policy, the Davidson School of Chemical Engineering has established procedures for students seeking accommodations. These can be found online at the ChE Undergrad Office website. Only those accommodation requests that conform to both University and ChE policy guidelines will be implemented.

Some important points from the CHE policy include: Please give letters of accommodation to me and your academic advisor (and not the graduate TA for your recitation section). If you have your letter at the start of the term, I strongly recommend you give it to me within the first two weeks of the semester. If your accommodation involves exam conditions, I strongly urge you to provide a minimum of one week notice to ensure that the accommodations requested are available.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and

mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies and Statements.

Mental Health/Wellness Statement

Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765) 494-6995 during and after hours, on weekends and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to visit the Basic Needs Program website for support or e-mail basicneeds@purdue.edu.

Use of Copyrighted Materials

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Thus, these materials cannot be posted online (e.g., Chegg, Course Hero, etc.). Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor. To obtain permission to sell or barter notes, the individual wishing to sell or barter the notes must be registered in the course or must be an approved visitor to the class. Course instructors may choose to grant or not grant such permission at their own discretion and may require a review of the notes prior to their being sold or bartered. If they do grant such permission, they may revoke it at any time, if they so choose.

Disclaimer

This syllabus is subject to change. If any change occurs, it will be announced in the class and/or posted on Brightspace.

1. **CHE 37800 – Heat and Mass Transfer**

2. **Credits and contact hours:**
   4 credits
   Lecture – 3 days per week at 50 minutes for 15 weeks.
   Laboratory and recitation – four 110-minute lab sessions, 11 75-minute recitation sessions over 15 weeks.

3. **Instructor's or course coordinator's name:**

4. **Textbook(s):**
   S. Middleman, *An Introduction to Mass and Heat Transfer*, John Wiley & Sons, Inc., 1998.

   a. Other supplemental materials:
   Y.A. Çengel *et al.*, *Heat and Mass Transfer: Fundamentals and Applications*, 5th edition, McGraw-Hill, 2015.
   Incropera *et al.*, *Fundamentals of Heat and Mass Transfer*, 6th edition, John Wiley & Sons, Inc., 2007.
   Welty *et al.*, *Fundamentals of Momentum, Heat, and Mass Transfer*, 5th edition, John Wiley & Sons, Inc., 2008.
   E.L. Cussler, *Diffusion: Mass Transfer in Fluid Systems*, 3rd edition, Cambridge University Press, 2009.

5. **Specific course information**
   a. Catalog description: Macroscopic and differential energy balances. Heat transfer coefficients for free and forced convection and phase change. Conductive and radiative heat transfer. Applications to heat transfer equipment design and compressible fluid flow. Macroscopic and differential species balances. Mass transfer coefficients and analogies. Mass transfer with and without chemical reaction. Mass transfer equipment design.
   b. Prerequisites: CHE 37700
   c. Course status: Required

6. **Specific goals for the course**
   a. Specific outcomes of instruction:
   - Identify and explain the three mechanisms of heat transfer: conduction, convection, and radiation (20500, 21100→); (→43500, 45000)
   - Apply microscopic and macroscopic thermal energy balances incorporating Fourier's Law for the solution of problems describing steady-state and transient conduction, including convective heat transfer (20500, 21100→); (→43500, 45000)
   - Utilize mass and thermal energy balances for the design of heat transfer equipment and the analysis of its operation (20500, 21100→); (→42000, 43500, 45000, 45600)
   - Identify and explain the two main mechanisms of mass transfer: convection and diffusion (20500, 21100, 37700→); (→43500, 45000)
   - Apply microscopic and macroscopic species mass balances incorporating Fick's Law for the solution of problems describing steady-state and transient diffusion, including relevant

rate equations and convective mass transfer (20500, 21100, 34800→); (→42000, 43500, 45000)

- Identify and apply the analogies between momentum, heat, and mass transfer (21100, 37700→); (→43500, 45000)
- Utilize mass balances and appropriate phase equilibrium relations for the design of continuous mass transfer equipment and the analysis of its operation (20500, 21100, 30600→); (→42000, 43500, 45000, 45600)
- Work professionally and ethically in teams to conduct heat and mass transfer-based laboratory experiments (37700→); (→43500, 45000)
- Effectively report results in written form and practice safety as an integral part of laboratory work (37700→); (→42000, 43500, 45000)

b. Student outcomes addressed by the course:

1) Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.
2) Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3) Communicate effectively with a range of audiences.
5) Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6) Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7) Acquire and apply new knowledge as needed, using appropriate learning strategies.

7. Topics

differential and macroscopic total mass, species, and thermal energy balances
mechanisms of heat and mass transfer
steady-state and transient conduction
convective heat transfer
thermal circuit analogy
heat transfer system design
steady-state and transient diffusion
heterogeneous and homogeneous reacting systems
convective mass transfer
mass transfer system design
momentum, mass, heat transfer analogies